

PSG COLLEGE OF TECHNOLOGY
DEPARTMENT OF COMPUTER APPLICATIONS
23MX26 - Java Programming Laboratory
Assessment on Exception Handling

Problem 1: Password Strength Validator (Custom Exception)

Problem Statement

Create a class `PasswordValidator` with method:

```
void validate(String password)
```

This method checks whether a password is strong.

Rules for a Valid Password

- Minimum length = 8
- Must contain at least:
 - One uppercase letter
 - One lowercase letter
 - One digit

Exception Conditions

Throw exceptions with **specific messages**:

- If `length < 8` → "Password too short"
- If no uppercase → "Password must contain at least one uppercase letter"
- If no lowercase → "Password must contain at least one lowercase letter"
- If no digit → "Password must contain at least one digit"

☞ (Encourage students to create a **custom exception class** like `WeakPasswordException`)

Input Format

Each line contains a password string.

Output Format

- Print "Valid Password" if all conditions satisfied
- Otherwise print exception message

Sample Input

```
Hello123
hello123
HELLO123
Hi12
```

Sample Output

```
Valid Password
java.lang.Exception: Password must contain at least one uppercase letter
java.lang.Exception: Password must contain at least one lowercase letter
java.lang.Exception: Password too short
```

Problem 2: ATM PIN Verification System (Limited Attempts + Exception)

Problem Statement

Create a class `ATM` with method:

```
void verifyPin(int enteredPin)
```

System Behavior

- Correct PIN is **1234**
- User gets **maximum 3 attempts**

Exception Conditions

- If incorrect PIN → throw exception: "Invalid PIN"
- If attempts exceed 3 → throw exception: "Card blocked due to multiple invalid attempts"

Additional Requirement

- Maintain attempt count using a **static variable**
- Once blocked, system should not allow further validation

Input Format

Each line contains an integer PIN.

Output Format

- If correct → "Access Granted"
- If wrong → print exception message

Sample Input

```
1111
```

2222
3333
1234

Sample Output

```
java.lang.Exception: Invalid PIN
java.lang.Exception: Invalid PIN
java.lang.Exception: Card blocked due to multiple invalid attempts
java.lang.Exception: Card blocked due to multiple invalid attempts
```